

CASE STUDY

Learning Portal Development

Creating a centralized digital learning experience

ROLE	PRIMARY TOOLS	FOCUS
Learning Technologist / Learning Experience Developer / eLearning Developer / Web Designer	PowerPoint • Storyline • JavaScript • xAPI • LRS	UX/UI • Learning Portals • Progress Tracking

The Challenge

Several learning programs needed a centralized digital experience that organized and linked program-related training content in one place. The goal was to create a more streamlined learner experience while reducing reliance on external development vendors.

My Approach

I designed and developed custom learning portals that served as a centralized home for each learning program. I began by creating custom UX/UI concepts and visual designs in PowerPoint to establish the look, feel, and navigation experience, then translated those concepts into functional portals using Articulate Storyline.

Rather than using Storyline only for traditional course development, I extended its capabilities with variables, triggers, JavaScript, xAPI, and an LRS to create a more application-like learner experience with persistent progress tracking. Storyline was chosen to streamline development and maintenance.

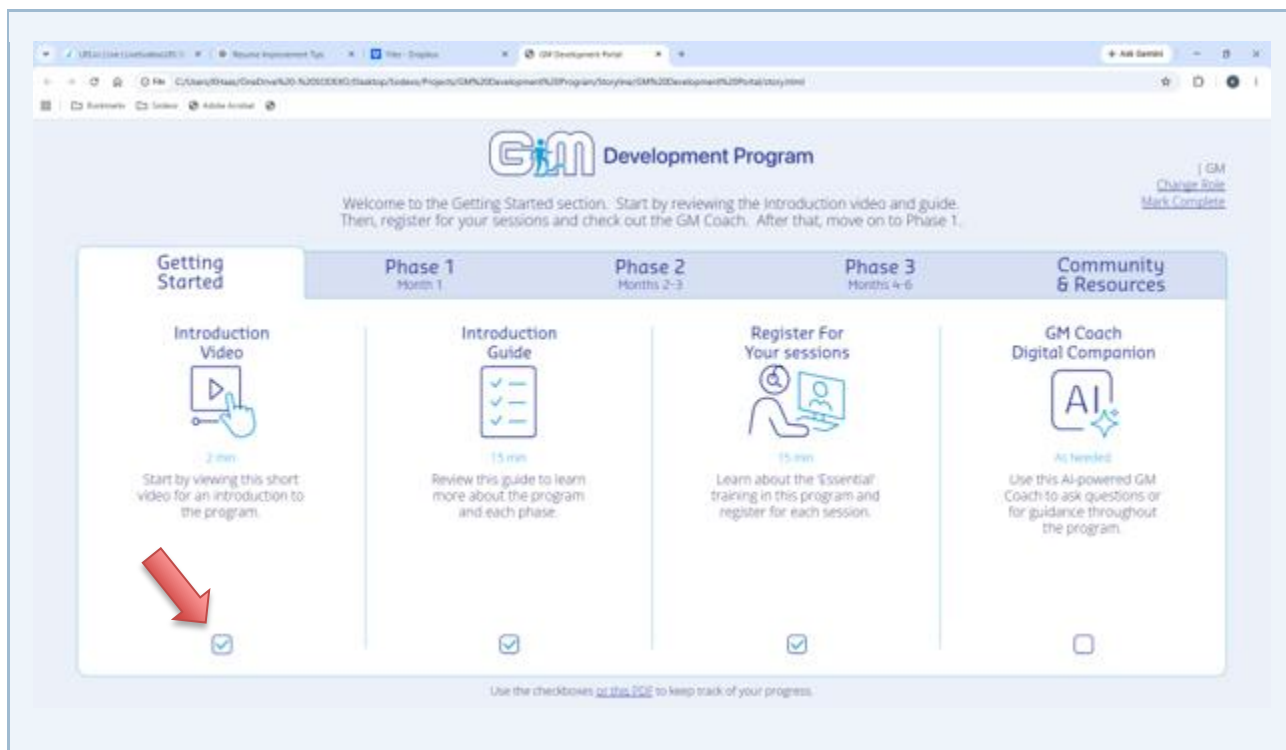
Technology & Development

ARTICULATE STORYLINE Used as the primary development platform for the portals, learner interface, navigation, interactions, and progress functionality.	JAVASCRIPT Extended Storyline capabilities and retrieved learner data from the LRS to maintain progress between browser sessions.
xAPI / LRS Implemented xAPI and an LRS to store learner activity and completion information and support reporting.	UX/UI DESIGN Used PowerPoint to create custom visual concepts and interfaces for individual programs to provide intuitive, cohesive learner experiences.

Learner Experience

One of the key features was persistent learner progress. Learners could visually identify which training activities they had completed using interactive checkboxes within the portal. Completion information was stored in the LRS and retrieved through JavaScript when learners returned to the portal, allowing their progress to persist across browser sessions.

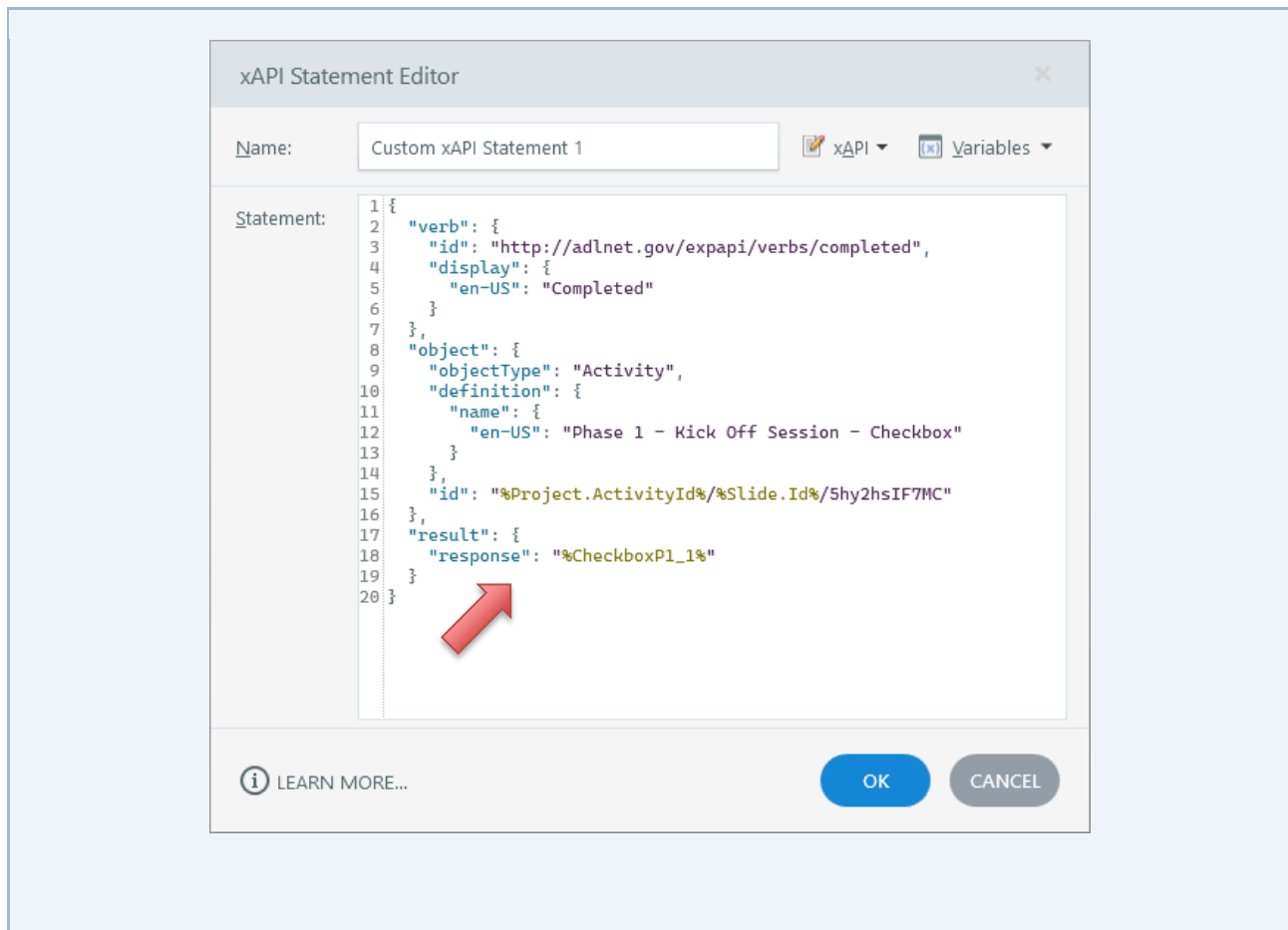
This allowed the portal to function as more than a collection of links—it became a centralized learning experience with built-in progress tracking.



Behind the Experience

LEARNER	STORYLINE PORTAL	xAPI	LRS	JAVASCRIPT → STORYLINE
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Conceptual flow: learner activity is captured through the portal, recorded through xAPI in the LRS, and retrieved by JavaScript so the Storyline experience can reflect persistent progress.



▼ [Icons] Kenneth L Haas Completed

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{
  "id": "f0f3e05a-6f6d-4d23-9a0f-341066577c65"
  "timestamp": "2026-08-14T16:06:59.229Z"
  "actor": {
    "objectType": "Agent"
    "account": {
      "name": "Kenneth L Haas"
      "homePage": "https://sodexo.csod.com"
    }
  }
  "verb": {
    "id": "http://adlnet.gov/expapi/verbs/completed"
    "display": {
      "en-US": "Completed"
    }
  }
  "result": {
    "response": "true"
  }
  "context": {...}
  "object": {
    "id": "https://sodexoelarning.azurewebsites.net/GMDDevelopmentProgramPortal/6MyRJ5yTqvF/6RPXWv37J"
    "objectType": "Activity"
    "definition": {
      "name": {
        "en-US": "Getting Started - Checkbox"
      }
    }
  }
}
  "stored": "2026-08-14T16:07:00.356Z"
  "authority": {...}
  "meta": {...}

```

JavaScript Editor

Object reference: But_P1B1 [ADD]

```

1 (async function() {
2   var checkboxMap = [
3     { objID: "6MyRJ5yTqvF/5hy2hsIF7MC", varName: "CheckboxP1_1" },
4     { objID: "6MyRJ5yTqvF/5LWnPCsfuwQ", varName: "CheckboxP1_2" },
5     { objID: "6MyRJ5yTqvF/6amPca9MjI3", varName: "CheckboxP1_3" },
6     { objID: "6MyRJ5yTqvF/6VXtFXG90MO", varName: "CheckboxP1_4" }
7   ];
8
9   for (const item of checkboxMap) {
10    try {
11      await getCheckbox(item.objID, item.varName);
12    } catch (xhr) {
13      console.error("Error fetching checkbox for", item.varName, xhr);
14    }
15  }
16})();

```

[i] LEARN MORE... [OK] [CANCEL]

Business Impact

- Brought learning-portal development in-house, reducing reliance on external vendors and saving the organization thousands of dollars in development costs.
- Created centralized, streamlined experiences for accessing program-related training content.
- Implemented persistent learner progress tracking using Storyline, JavaScript, xAPI, and an LRS.
- Supported learner activity analysis through custom LRS reporting dashboards.

What I Learned

This work expanded my role beyond traditional eLearning or Articulate Storyline development and reinforced the value of combining learning design with technology, UX/UI, and data. It demonstrated that Articulate Storyline can support more than traditional eLearning when paired with JavaScript, xAPI, and an LRS.